

Nir Bitansky - Homepage

I'm on leave, and have moved to NYU, my updated webpage is [here](#).

I'm a faculty member at [Tel Aviv University's School of Computer Science](#). I'm interested in the theory of computation at large, and the theory of cryptography in particular.

Before joining TAU I was a postdoctoral researcher at MIT CSAIL hosted by [Vinod Vaikuntanathan](#). I completed my Ph.D. at Tel Aviv University where I was advised by [Ran Canetti](#).

Contact

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[Students and Postdocs](#)[Publications](#)[Some Videotaped Talks](#)[Program Committees](#)[Graduate Teaching](#)[Undergraduate Teaching](#)

Students and Postdocs

Ph.D.

Sapir Freizeit

Nathan Geier

[Omri Shmueli](#) (co-advised with Zvika Brakerski)

M.Sc.

Dana Shamir (co-advised with Omer Paneth)

Tomer Solomon (graduated 2023)

Noa Eizenstadt (graduated 2020)

Omri Shmueli (graduated 2020)

Nathan Geier (graduated 2021)

Michael Kelner (graduated 2021)

Idan Gerichter (graduated 2022)

Sapir Freizeit (graduated 2022)

Postdoc

Chethan Kamath (co-hosted with Omer Paneth, now Assistant Professor at IIT Bombay)

Eylon Yogev (co-hosted with Omer Paneth, now Assistant Professor at Bar-Ilan University)

Publications

Batch Proofs are Statistically Hiding

with Chethan Kamath, Omer Paneth, Ron Rothblum, and Prashant Vasudevan

STOC 2024 [PDF](#)

Non-Interactive Universal Arguments



with Omer Paneth, Dana Shamir, and Tomer Solomon

CRYPTO 2023 [PDF](#)

Bootstrapping Homomorphic Encryption via Functional Encryption

with Tomer Solomon

ITCS 2023 [PDF](#)

PPAD is as Hard as LWE and Iterated Squaring

with Arka Rai Choudhuri, Justin Homgren, Chethan Kamath, Alex Lombardi, Omer Paneth, and Ron D. Rothblum

TCC 2022 [PDF](#)

Constructive Post-quantum Reductions

with [Zvika Brakerski](#) and [Yael Tauman Kalai](#)

CRYPTO , QCRYPT 2022 [PDF](#)

Statistically Sender Private OT from LPN and Derandomization

with Sapir Freizeit

CRYPTO 2022 [PDF](#)

Non-malleable Commitments Against Quantum Attacks

with [Huijia Lin](#) and Omri Shmueli

EUROCRYPT, QCRYPT 2022 [PDF](#)

Classical Binding for Quantum Commitments

with [Zvika Brakerski](#)

TCC 2021 [PDF](#)

Post-quantum Resetably-Sound Zero Knowledge

with Michael Kelner and Omri Shmueli

 C 2021, QCRYPT 2021 [PDF](#)

Characterizing Deterministic Prover Zero Knowledge

with [Arka Rai Choudhuri](#)

TCC 2020 [PDF](#)

Weakly Extractable One Way Functions

with Noa Eizenstadt and [Omer Paneth](#)

TCC 2020 [PDF](#)

Post-quantum Zero Knowledge in Constant Rounds

with Omri Shmueli

STOC 2020, QCRYPT 2020, QIP 2021 [PDF](#)

On the Cryptographic Hardness of Local Search

with Idan Gerichter

ITCS 2020 [PDF](#)

On Oblivious Amplification of Coin-Tossing Protocols

with Nathan Geier

ITCS 2020 [PDF](#)

On the Complexity of Collision Resistant Hash Functions: New and Old Black-Box Separations

with Akshay Degwekar

TCC 2019 [PDF](#)

On Round-Optimal Statistical Zero-Knowledge Arguments

with [Omer Paneth](#)

CRYPTO 2019 [PDF](#)

Weak Zero-Knowledge Beyond the Black-Box Barrier

 with [Dakshita Khurana](#) and [Omer Paneth](#)

STOC 2019, Siam Journal on Computing 2022 [PDF](#)

Distributional Collision Resistance Beyond One-Way Functions

with [Iftach Haitner](#), [Ilan Komargodski](#), and [Eylon Yogev](#)

EUROCRYPT 2019 [PDF](#)

One-Message Zero Knowledge and Non-Malleable Commitments

with [Huijia Lin](#)

TCC 2018 [PDF](#)

Multi-Collision Resistance: A Paradigm for Keyless Hash Functions

with [Yael Tauman Kalai](#), and [Omer Paneth](#)

STOC 2018 [PDF](#)

Verifiable Random Functions from Non-Interactive Witness-Indistinguishable Proofs

TCC 2017, Journal of Cryptology 2019 [PDF](#)

Structure vs Hardness through the Obfuscation Lens

with Akshay Degwekar and [Vinod Vaikuntanathan](#)

CRYPTO 2017, Siam Journal on Computing 2021 [Draft](#)

On Removing Graded Encodings from Functional Encryption

with [Huijia Lin](#) and [Omer Paneth](#)

EUROCRYPT 2017 [PDF](#)

A Note on Perfect Correctness by Derandomization

with [Vinod Vaikuntanathan](#)

 JROCRYPT 2017, Journal of Cryptology 2022 [PDF](#)

From Cryptomania to Obfuscation through Secret-Key Functional Encryption

with [Ryo Nishimaki](#), [Alain Passelègue](#), and [Daniel Wichs](#)

TCC 2016B, Journal of Cryptology 2019 [PDF](#)

3-Message Zero Knowledge against Human Ignorance

with [Zvika Brakerski](#), [Yael Tauman Kalai](#), [Omer Paneth](#), and [Vinod Vaikuntanathan](#)

TCC 2016B [PDF](#)

Time-Lock Puzzles from Randomized Encodings

with [Shafi Goldwasser](#), [Abhishek Jain](#), [Omer Paneth](#), [Vinod Vaikuntanathan](#), and [Brent Waters](#)

ITCS 2016 [PDF](#)

Indistinguishability Obfuscation: from Approximate to Exact

with [Vinod Vaikuntanathan](#)

TCC 2016A [PDF](#)

Perfect Structure on the Edge of Chaos

with [Omer Paneth](#) and [Daniel Wichs](#)

TCC 2016A [PDF](#)

Indistinguishability Obfuscation from Functional Encryption

with [Vinod Vaikuntanathan](#)

FOCS 2015, Journal of the ACM 2018 [PDF](#)

On the Cryptographic Hardness of Finding a Nash Equilibrium

with [Omer Paneth](#) and [Alon Rosen](#)

FOCS 2015 [PDF](#)

Succinct Randomized Encodings and their Applications

with [Sanjam Garg](#), [Huijia Lin](#), [Rafael Pass](#), and [Sidharth Telang](#)

STOC 2015 [PDF](#), Siam Journal on Computing 2019 (merged with Canetti, Holmgren, Jain, and Vaikuntanathan)

Merge of [\[BGT\]](#) and [\[LP\]](#)

ZAPs and Non-Interactive Witness-Indistinguishability from Indistinguishability Obfuscation

with [Omer Paneth](#)

TCC 2015 [PDF](#)

On Virtual Grey Box Obfuscation for General Circuits

with [Ran Canetti](#), [Yael Tauman Kalai](#), and [Omer Paneth](#)

CRYPTO 2014, Algorithmica 2016 [Draft](#)

The Impossibility of Obfuscation with Auxiliary Input or a Universal Simulator

with [Ran Canetti](#), [Henry Cohn](#), [Shafi Goldwasser](#), [Yael Tauman Kalai](#), [Omer Paneth](#), and [Alon Rosen](#)

CRYPTO 2014 [PDF](#)

Merge of [\[BCPR\]](#) and [\[CGK\]](#)

Leakage-Tolerant Computation with Input-Independent Preprocessing

with [Dana Dachman-Soled](#) and [Huijia Lin](#),

CRYPTO 2014 [PDF](#)

On the Existence of Extractable One-Way Functions

with [Ran Canetti](#), [Omer Paneth](#), and [Alon Rosen](#)



STOC 2014, Siam Journal on Computing 2016 [Draft](#)

Merge of [\[BCP\]](#) and [\[BCPR\]](#)

Obfuscation for Evasive Functions

with [Boaz Barak](#), [Ran Canetti](#), [Yael Tauman Kalai](#), [Omer Paneth](#), and [Amit Sahai](#)

TCC 2014 [PDF](#)

Why "Fiat-Shamir for Proofs" Lacks a Proof

with [Dana Dachman-Soled](#), [Sanjam Garg](#), [Abhishek Jain](#), [Yael Tauman Kalai](#), [Adriana Lopez-Alt](#), and [Daniel Wichs](#)

TCC 2013

Merge of [\[BGW\]](#) and [\[DJKL\]](#)

Succinct Non-Interactive Arguments via Linear Interactive Proofs

with [Alessandro Chiesa](#), [Yuval Ishai](#), [Rafail Ostrovsky](#), [Omer Paneth](#)

TCC 2013, Journal of Cryptology 2022 [PDF](#)

Recursive Composition and Bootstrapping for SNARKs and Proof-Carrying Data

with [Ran Canetti](#), [Alessandro Chiesa](#), [Eran Tromer](#)

STOC 2013 [PDF](#)

On the Impossibility of Approximate Obfuscation and Applications to Resettable Cryptography

with [Omer Paneth](#)

STOC 2013 (subsumed by Siam Journal on Computing 2015 [Draft](#))

From the Impossibility of Obfuscation to a New Non-Black-Box Simulation Technique

with [Omer Paneth](#)



FOCS 2012, Siam Journal on Computing 2015 [Draft](#)

Succinct Arguments from Multi-Prover Interactive Proofs and their Efficiency Benefits

with [Alessandro Chiesa](#)

CRYPTO 2012 [PDF](#)

Leakage-Tolerant Interactive Protocols

with [Ran Canetti](#) and [Shai Halevi](#)

TCC 2012 [PDF](#)

Point Obfuscation and 3-Message Zero Knowledge

with [Omer Paneth](#)

TCC 2012 [PDF](#)

The Hunting of the SNARK

with [Ran Canetti](#), [Alessandro Chiesa](#), [Shafi Goldwasser](#), [Huijia Lin](#), [Aviad Rubinstein](#), [Eran Tromer](#)

Journal of Cryptology 2016 [PDF](#)

Preliminary version: From Extractable Collision Resistance to Succinct Non-Interactive Arguments of Knowledge, and Back Again

with [Ran Canetti](#), [Alessandro Chiesa](#), [Eran Tromer](#)

ITCS 2012

Program Obfuscation with Leaky Hardware

with [Ran Canetti](#), [Shafi Goldwasser](#), [Shai Halevi](#), [Yael Tauman Kalai](#), [Guy N. Rothblum](#)

ASIACRYPT 2011 [PDF](#)

On Strong Simulation and Composable Point Obfuscation

 with [Ran Canetti](#)

CRYPTO 2010, Journal of Cryptology 2013 [PDF](#)

Thesis

Getting Inside the Adversary's Head: New Directions in Non-Black-Box Knowledge Extraction

Tel Aviv University 2015 [PDF](#)



Some Videotaped Talks

[Constructive Post-quantum Reductions](#) (Crypto '22)

[Classical Binding for Quantum Commitments](#) (TCC '21)

[On the Cryptographic Hardness of Local Search](#) (ITCS '20)

[On Round-Optimal Statistical Zero-Knowledge Arguments](#) (Crypto '19)

[Our Current Knowledge of Knowledge Assumptions](#) (Dimacs workshop on cryptographic assumptions '17)

[On Removing Graded Encodings from Functional Encryption](#) (Eurocrypt '17)

[A Note on Perfect Correctness by Derandomization](#) (Eurocrypt '17)

[A Reduction from Flying Pigs to Jumping Pigs](#) (IAS, March '17)

[On the Cryptographic Hardness of Finding a Nash Equilibrium](#) (IAS, March '17, [shorter version](#) at FOCS '15)

[Program Obfuscation: The Power of Unreadable Code](#) (UW Colloquium, April '16)

[Indistinguishability Obfuscation from Functional Encryption](#) (FOCS '15)

[Time-Lock Puzzles from Randomized Encodings](#) (Simons Institute, July '15)

[On Virtual Grey Box Obfuscation for General Circuits](#) (Crypto '14)

[Leakage-Tolerant Computation with Input-Independent Preprocessing](#) (Crypto '14)

Program Committees

Crypto	2024 (Area Chair) , 2021
Eurocrypt	2021 , 2019 , 2018 , 2017
Foundations of Computer Science (FOCS)	2022 , 2018
Innovations in Theoretical Computer Science (ITCS)	2023 , 2020
Public Key Cryptography (PKC)	2017
Theory of Cryptography Conference (TCC)	2022 , 2019 , 2017 , 2016A , 2015



Graduate Teaching

Foundations of Cryptography	Fall 2022, Fall 2021, Fall 2019 , Fall 2017
Foundations of Cryptography: Advanced Topics	Spring 2023, Spring 2021
Lattices	Fall 2021, Fall 2019
Seminar on Cryptographic Protocols	Fall 2021
Seminar on Encrypted Computation	Spring 2020
Seminar on Zero-Knowledge Proofs	Spring 2019
Seminar on Advanced Topics in Crypto	Spring 2022

Undergraduate Teaching

The Theory Behind Blockchains	Spring 2019
Computational Models	Spring 2020-2024

